a1852681
Nicholas Corker

Written Self-Assessment
Project developed: MATLAB Wordle

My MATLAB project demonstrates excellent conceptual coverage. I demonstrate all criterial outlined within the task for conceptual coverage within the program. I use multiple unique functions, vectors and matrices, strings, while and for loops, image display and creation, and conditional, is in the program. Function usage is present within the selection of the word (selectWord) and the comparison between the user guesses and the word the user must guess (searchCorrectPositon, searchIncorrectPosition). The output of the latter function is a matrix. The main game will only continue under the conditional execution of a while loop, where the number of guesses the user has had does not exceed the maximum number of guesses.

The functionality of my program is excellent. I use a variety of new functions and functionality within my program including the 'cat' command in my imageDisplay function, which concatenates the newly generated image vertically, underneath the previously generated images. All external contributions to the functionality of the program are from the Mathworks MATLAB documentation and the Mathworks discussion forums, as I worked to ensure that I was not taking parts of code from solutions found across the internet, and intended to work on my own solutions, and solving errors using only MATLAB documentation. Any additional sources that I did use are clearly detailed and referenced.

I show substantial development of the MATLAB program. Initially I attempted to create the program all in one .m file, titled wordleMatlabVersionV1, but instead opted to create individual functions, testing them individually, before putting them all together in my final version. The incremental development is not perfect, as it would have been better to save a new version of the main program each time that I successfully incorporated a new, working function.

The testing process for my functions is excellent. Within the testing drivers, I created several test cases which cover all possible scenarios and areas where I would expect there to be an error. This includes in the searchCorrectPosition function, where I put in a string with more characters than expected, which case due to the defined range of the loop, it did not operate on the letter that fell outside the standard string length.

My project shows substantial use of comments and overall good coding style. There are comments for every critical line in the code, and there are also additional comments which are used to inform what specific loops and functions do. The comments are also present in the individual functions and test drivers. The style does lack occasionally as I do not always follow the 80-column rule in my code, often going beyond 120 characters. This does mean that the code can be difficult to read, but generally beyond 80 characters, I am only printing additional comments. I consistently use good indentation to give a clear indication of variables, the presence of loops, and the visibility of elements and code contained within the loops.